

MambaNA: A state space model with Neighborhood Attention for Medical Super-Resolution

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Abstract—In recent years, Mamba-based methods have demonstrated significant potential in various tasks, including medical image super-resolution, due to Mamba's ability to balance global receptive fields and computational efficiency. However, the inherent causal modeling of Mamba restricts its ability to capture local features, which poses a challenge for medical image super-resolution, where intricate local details in structures such as cells and tissue boundaries are crucial. In this work, we propose MambaNA, a novel medical image super-resolution method that integrates a neighborhood attention mechanism into the Mamba model. Specifically, MambaNA computes pixel-wise attention weights for neighboring elements to introduce non-causality while extracting local features that maintain translational equivariance. Furthermore, it performs multi-frequency channel fusion directly following the SSM's sequence modeling, which significantly improves model performance and training stability. Extensive experiments demonstrate the superior performance of MambaNA in medical image super-resolution tasks.

Keywords—Medical image super-resolution; Mamba; Neighborhood Attention; State Space Model

I. INTRODUCTION

Modern medical diagnosis relies heavily on the visual information provided by medical imaging. However, acquiring high-quality images often requires expensive advanced equipment and prolonged patient exposure. Recent advances in deep learning have made super-resolution (SR) a cost-effective solution for image enhancement, enabling the reconstruction of fine structures from low-resolution inputs. In the medical domain, SR aims to generate high-resolution outputs from low-resolution scans by recovering details that are typically lost during image acquisition.

Early super-resolution methods, based on simple interpolation [1]–[3] and optimization [4], were straightforward but often failed to faithfully restore fine details. Subsequently, Convolutional Neural Networks (CNNs) [5]–[10] and Generative Adversarial Networks (GANs) [9], [11], [12] were widely adopted because of their ability to learn nonlinear mappings and rich feature representations. However, these methods often struggle to capture long-range dependencies and global receptive fields, which limits their ability to restore high-frequency details and maintain structural coherence, especially in complex medical images.

To address these limitations, the Transformer architecture [13], with its powerful self-attention mechanism for

global dependency modeling, has been successfully adapted to computer vision, notably through the Vision Transformer (ViT) [14]–[18]. Although effective at capturing long-range spatial relationships, Transformer-based methods often incur substantial computational and memory costs because of the quadratic complexity of global self-attention. State Space Models (SSMs) [19]–[21] have recently emerged as efficient sequence models for long-range dependency modeling. Among them, Mamba [22] is particularly attractive because it balances global receptive fields with computational efficiency. However, its inherent causal modeling limits its ability to fully exploit local features [23], which is especially problematic for medical images, where fine localized details are crucial for accurate analysis.

To address these limitations, we propose MambaNA, a novel medical image super-resolution method. It integrates Neighborhood Attention (NA) [24] into a Mamba architecture to introduce non-causal modeling and alleviate Mamba's inherent causal limitations while enhancing local feature extraction. In addition, MambaNA incorporates a lightweight Channel Mixing module, implemented with multi-spectral channel attention (MSCA) [25] based on the Discrete Cosine Transform (DCT). This design captures intricate frequency-domain patterns and cross-channel dependencies, and experimental results show that it significantly improves both performance and training stability. The main contributions of this paper are summarized as follows:

- 1) We develop MambaNA, a Mamba-based network for medical image super-resolution that combines efficient global modeling with non-causal local attention for more accurate image reconstruction.
- 2) We propose an SSM-Layer equipped with a Channel Mixing module directly after the SSM module. Implemented as a multi-spectral channel attention (MSCA) design, it adaptively weights frequency components within channels, thereby improving both reconstruction quality and training stability.
- 3) We conduct extensive experiments and ablation studies to validate the contribution of each component. The results show that MambaNA achieves strong and consistent performance across multiple super-resolution benchmarks.

II. RELATED WORKS

A. Medical Image Super-Resolution

Deep learning has substantially advanced medical image SR. CNN-based methods such as SRCNN and VDSR, together with residual and dense variants [6], [26]-[30], have improved detail recovery across CT and MRI. GAN-based methods further enhance perceptual quality in low-dose CT and MRI reconstruction [31]-[34]. More recently, Transformer-based models have introduced stronger global context modeling for SR [35]-[38], although their memory and computational costs remain high for high-resolution medical images.

B. State Space Models and Attention

SSMs provide an efficient alternative for long-range dependency modeling. Mamba [22] and its vision variants [23], [39]-[43] improve global modeling efficiency, yet their causal scanning process is less suitable for dense 2D local interactions. In parallel, attention mechanisms remain effective for refining feature representations. Neighborhood Attention preserves local inductive bias and translation equivariance in spatial modeling [24], while frequency-aware channel attention such as FcaNet [25] improves channel discrimination. These observations motivate our design: combining Mamba for efficient global context modeling with local spatial attention and frequency-aware channel modulation for medical SR.

III. METHODOLOGY

A. Preliminaries

1) State Space Models

SSMs model an input sequence x and output sequence y through a hidden state h . A continuous-time SSM is written as:

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t), \\ y(t) &= Ch(t) + Dx(t). \end{aligned} \quad (1)$$

and discretized for neural sequence modeling as:

$$\begin{aligned} h_t &= Ah_{t-1} + Bx_t, \\ y(t) &= Ch_t + Dx_t. \end{aligned} \quad (2)$$

Mamba [22] extends this formulation with input-dependent parameters, enabling efficient long-range dependency modeling with linear complexity.

B. Overall Architecture

As shown in Figure 1, MambaNA consists of shallow feature extraction, deep feature extraction, and image reconstruction modules. Given an input image I_{LQ} , a 3×3 convolution first produces shallow features F_0 . Deep features are then extracted by stacking N_1 residual MambaNA groups, each containing N_2 MambaNA blocks. Each block is composed of an Attention Block and an SSM-Layer. The refined features are finally used to reconstruct the high-quality output image.

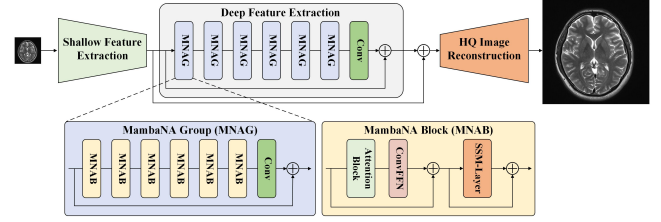


Figure 1. Overall architecture of MambaNA for medical image super-resolution. The network consists of shallow feature extraction, stacked MambaNA groups for deep feature modeling, and a final reconstruction module for generating the high-resolution output.

For medical image super-resolution, we optimize MambaNA with a hybrid loss:

$$\mathcal{L} = 0.7 \cdot \mathcal{L}_{L-1} + 0.3 \cdot \mathcal{L}_{\text{perceptual}}, \quad (3)$$

$$\mathcal{L}_{\text{perceptual}} = \|\phi(I_{HQ}) - \phi(I_{GT})\|_2^2,$$

where $\phi(\cdot)$ denotes feature extraction from pre-trained VGG-19's 5th conv layer.

C. Attention Block

As illustrated in Figure 2, MambaNA adopts Neighborhood Attention (NA) [24] to introduce non-causal local modeling. Unlike global or window-based attention, NA restricts each query to a local neighborhood, making it well suited to preserving subtle anatomical structures. Given an input $X \in \mathbb{R}^{N \times d}$, the attention at position i is computed as

$$\begin{aligned} A_i^k &= Q_i k_{N(i)}^T + B_i, \\ NA_k(i) &= \text{softmax}\left(\frac{A_i^k}{\sqrt{d}}\right) V_{N(i)}. \end{aligned} \quad (4)$$

This design improves fine-grained local interaction while preserving translation equivariance, complementing Mamba's global modeling.

D. SSM-Layer

As shown in Figure 3, the SSM-Layer captures long-range dependencies through adaptive pixel reordering. Given a feature map $X \in \mathbb{R}^{C \times H \times W}$, a semantic relevance score is estimated for each pixel:

$$s(i) = \text{MLP}(\text{GAP}(X_i)), \quad (5)$$

and quantized into G groups:

$$g(i) = \lfloor G \cdot \sigma(s(i)) \rfloor. \quad (6)$$

Pixels are then concatenated by group to form the scan order:

$$\pi = \text{concat}(\pi_{g=0}, \pi_{g=1}, \dots, \pi_{g=G-1}). \quad (7)$$

This adaptive sequence reduces semantic distance within the SSM scan while keeping linear complexity.

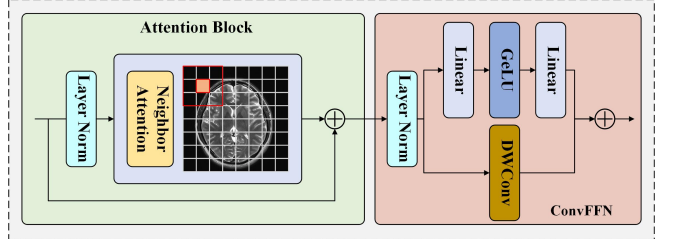


Figure 2. Structure of the Attention Block in MambaNA. It applies Neighborhood Attention to strengthen local non-causal interactions and improve the reconstruction of fine anatomical details.

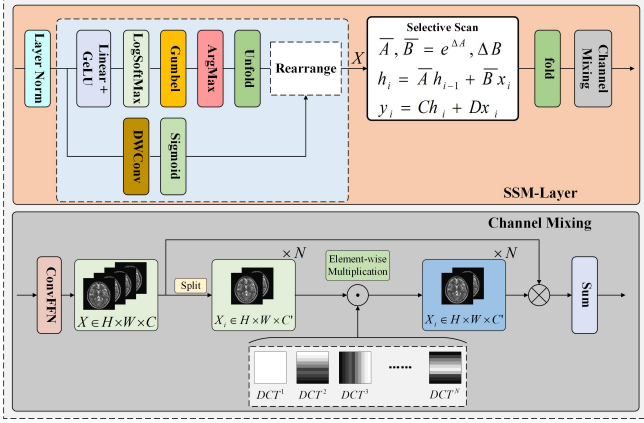


Figure 3. Structure of the proposed SSM-Layer in MambaNA. The layer combines adaptive scan ordering with multi-spectral channel attention to capture global dependencies and refine frequency-aware channel responses.

To further refine channel responses, we apply a Channel Mixing module after the SSM module. Implemented with multi-spectral channel attention (MSCA), this module uses fixed DCT bases to aggregate frequency responses and produce channel weights:

$$\begin{aligned} \text{Freq}^i &= \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} X^{i,h,w} \cdot \text{DCT}^{i,h,w}, \\ \text{Freq} &= \text{cat}([\text{Freq}^1 \text{Freq}^2 \dots \text{Freq}^C]), \\ w &= \sigma(\text{MLP}(\text{Freq})). \end{aligned} \quad (8)$$

The Channel Mixing module strengthens frequency-aware channel modulation and improves optimization stability.

IV. EXPERIMENTS

A. Experimental Settings

1) Datasets and Metrics

Our experiments utilized two publicly available datasets for medical image super-resolution: the IXI dataset¹ and a Brain Tumor MRI dataset². The IXI dataset includes 3D multimodal MRI scans with T1, T2, and PD modalities. We focused on the T1-weighted subset (IXI-T1) for SR training and evaluation because of its superior soft-tissue contrast. Each original IXI-T1 scan contains 96 slices of size 240×240 pixels; we uniformly selected 8 representative slices per scan to avoid redundant low-variation samples. The IXI-T1 dataset was split as follows: 600 slices for training, 64 slices for validation, and 84 slices from remaining qualified T1 scans for testing. All slices retain the 240×240 pixel resolution.

To quantitatively evaluate the quality of the super-resolved images, we adopted two widely used metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). PSNR measures pixel-wise fidelity by comparing reconstructed images with the ground truth, while SSIM evaluates perceptual similarity in terms of luminance, contrast, and structure.

¹ <https://brain-development.org/ixi-dataset/>

² <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

2) Experimental Details

Our MambaNA model was implemented in PyTorch and trained on an NVIDIA RTX A6000 GPU. The low-resolution training samples were generated in the frequency domain to better match the distribution of real low-resolution images. We used Adam as the optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The initial learning rate and batch size were set to 2×10^{-4} and 2, respectively. Additionally, the original images were cropped into 64×64 patches during training. We used six Mamba blocks, each with 192 channels.

TABLE I. QUANTITATIVE COMPARISON OF MEDICAL IMAGE SUPER-RESOLUTION PERFORMANCE.

Method	Scale	Brain Tumor		IXI-T1	
		PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
Bicubic(2D)	2 $\times$	29.96	0.8397	27.57	0.8523
EDSR[44]	2 $\times$	31.70	0.8931	32.08	0.9154
SwinIR[16]	2 $\times$	34.60	0.9047	31.82	0.9031
HAT[46]	2 $\times$	34.94	0.9142	32.16	0.9284
MambaIR[43]	2 $\times$	35.75	0.9362	33.30	0.9650
MambaIRv2[23]	2 $\times$	36.43	0.9431	33.58	0.9664
MambaNA[Ours]	2 $\times$	36.95	0.9447	33.94	0.9686
Bicubic(2D)	4 $\times$	18.85	0.7535	21.22	0.8293
EDSR[44]	4 $\times$	28.10	0.9059	28.45	0.8925
SwinIR[16]	4 $\times$	30.23	0.9035	28.12	0.9011
HAT[46]	4 $\times$	30.53	0.9101	28.56	0.9054
MambaIR[43]	4 $\times$	30.75	0.9202	29.56	0.9073
MambaIRv2[23]	4 $\times$	31.50	0.9320	30.07	0.9234
MambaNA[Ours]	4 $\times$	32.28	0.9386	30.64	0.9279

Our evaluation first focused on comparing MambaNA with existing medical image super-resolution methods on the Brain Tumor and IXI-T1 datasets.

As detailed in TABLE I, MambaNA consistently achieves higher PSNR and SSIM values than [45], [23], and [16]. We additionally include HAT as a strong Transformer baseline for comparison.

These quantitative results highlight MambaNA's strong ability to reconstruct high-fidelity medical images while balancing global context modeling and local detail recovery.

Beyond numerical metrics, Figure 4 presents a qualitative comparison of the super-resolution results. The first row shows the reconstructed images from different methods, while the second row presents the corresponding pixel-wise mean squared error (MSE) maps with respect to the ground truth. Darker regions in the error maps indicate smaller deviations from the ground truth and therefore better reconstruction fidelity.

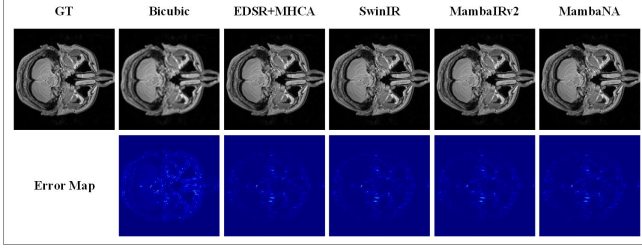


Figure 4. Qualitative comparison of reconstruction results and corresponding error maps. Darker regions indicate smaller reconstruction errors, showing that MambaNA preserves anatomical structures more faithfully than competing methods.

As shown in the figure, MambaNA produces reconstructions that are visually closer to the ground truth, with fewer artifacts and better preservation of fine anatomical structures than competing methods. This qualitative superiority further supports the practical value of MambaNA for clinical image analysis.

To assess the generalization capability of MambaNA, we also evaluated it on widely used natural image super-resolution benchmarks, including BSDS100 and Urban100. As presented in TABLE II, MambaNA consistently achieves competitive and often best or near-best PSNR and SSIM results across these datasets, demonstrating strong competitiveness against established natural image super-resolution methods. These results indicate that, although MambaNA is designed for medical images, its super-resolution capability transfers effectively to diverse natural image scenarios, highlighting its versatility and broad applicability.

TABLE II. QUANTITATIVE COMPARISON ON CLASSICAL NATURAL IMAGE SUPER-RESOLUTION BENCHMARKS.

Method	Scale	BSDS100		Urban100	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑
EDSR[44]	2×	32.23	0.9013	32.93	0.9351
SwinIR[16]	2×	31.33	0.9022	32.98	0.9363
HAT[46]	2×	31.25	0.9035	33.54	0.9387
MambaIR[43]	2×	31.98	0.9042	34.06	0.9446
MambaIRv2[23]	2×	32.71	0.9046	34.88	0.9471
MambaNA[Ours]	2×	32.82	0.9054	35.36	0.9493
EDSR[44]	4×	26.68	0.7298	26.63	0.8012
SwinIR[16]	4×	26.78	0.7382	27.39	0.8213
HAT[46]	4×	26.90	0.7429	27.42	0.8255
MambaIR[43]	4×	27.90	0.7489	27.51	0.8273
MambaIRv2[23]	4×	27.86	0.7498	27.72	0.8335
MambaNA[Ours]	4×	27.93	0.7488	28.04	0.8511

B. Ablation Study

We conducted ablation experiments to evaluate the contributions of the Attention Block and the Channel Mixing module under identical training settings. Specifically, we

considered three ablated variants by removing the Attention Block, removing the Channel Mixing module, and removing both modules simultaneously. All variants were trained and evaluated using the same dataset split, optimizer, learning rate schedule, and number of epochs to ensure a fair comparison.

TABLE III. ABLATION RESULTS FOR THE MAIN COMPONENTS OF MAMBA NA.

Attention Block	Channel Mixing	PSNR↑	SSIM↑
√	√	35.45	0.9567
×	√	34.41	0.9413
√	×	34.65	0.9462
×	×	33.68	0.9274

As reported in TABLE III, the full MambaNA model achieves the best average PSNR and SSIM across the two medical datasets. Removing the Attention Block reduces PSNR from 35.45 dB to 34.41 dB and SSIM from 0.9567 to 0.9413, while removing the Channel Mixing module reduces PSNR to 34.65 dB and SSIM to 0.9462. When both modules are removed, performance drops further to 33.68 dB PSNR and 0.9274 SSIM. These results confirm that both modules contribute positively to reconstruction quality, and that their combination yields the strongest overall performance. In addition, we observed that the Channel Mixing module helps improve training stability during optimization.

V. CONCLUSIONS

In this paper, we developed MambaNA, a novel super-resolution framework tailored for medical image reconstruction. By adopting the Mamba architecture, the model benefits from efficient long-range dependency modeling, which is important for processing high-resolution medical data. To overcome the limitations of Mamba's causal sequence modeling in 2D image restoration, MambaNA integrates both an Attention Block and a Channel Mixing module. The Attention Block introduces non-causal local modeling into the Mamba backbone, while the Channel Mixing module improves training stability and feature refinement. Extensive experiments demonstrated MambaNA's strong performance on medical image super-resolution tasks and highly competitive generalization on natural image benchmarks. The ablation results further show that combining both modules yields the best average PSNR and SSIM across the two medical datasets, confirming the effectiveness of the overall design. This work underscores the significant potential of Mamba-based architectures in high-fidelity image reconstruction. Future work will focus on improving computational efficiency for extremely large 3D volumes and exploring unsupervised learning strategies to reduce data dependency.

ACKNOWLEDGMENT

This work was supported by the Natural Science Foundation of Guangdong Province under Grant No. 2024A1515010255.

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